

Feasible cricket research



which is feasible research on cricket in year also for mtech student of easily available data

Good question 👍. For an **M.Tech research project in cricket**, the best choice is something where **data is easily available, research scope is strong, and feasibility is high**.

Here are some **feasible research areas in cricket with easily available data**:

♦ 1. Performance Analytics (Player & Team)

- **Idea:** Analyze batting, bowling, and fielding performance using open datasets.
- **Data Sources:**
 - [ESPN Cricinfo StatsGuru](#) ↗
 - [Cricbuzz match data]
 - Kaggle cricket datasets
- **Research Topics:**
 - Predicting match outcomes (classification)
 - Player ranking system using ML (similar to ICC ratings but data-driven)
 - Impact of strike rate vs. average in T20 matches
- **Feasibility:** ✅ Easy (data-rich, well-studied, many datasets available).

♦ 2. Cricket Match Outcome Prediction

- **Idea:** Use machine learning to predict win probability during a match.
- **Data Sources:** Ball-by-ball datasets (Kaggle, Cricmetric, Cricsheet).
- **Research Topics:**
 - T20 match outcome prediction after each over
 - Impact of toss & pitch conditions on results
 - Deep learning for live win probability (similar to WinViz in cricket broadcasts).
- **Feasibility:** ✅ Very popular & many datasets exist.

♦ 3. Injury & Workload Analysis (Sports Science + AI)

- **Idea:** Study workload patterns to reduce injuries.
 - **Data Sources:** Player workload (overs bowled, match frequency, travel schedules from Cricinfo).
 - **Research Topics:**
 - Bowling workload → injury risk prediction
 - Scheduling optimization for players
 - **Feasibility:** ⚠ Medium (needs curated data, not always open).
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♦ 4. Decision Review System (DRS) & Computer Vision

- **Idea:** AI for umpiring assistance.
 - **Data Sources:** Match videos, open datasets of cricket highlights (YouTube, Kaggle).
 - **Research Topics:**
 - Ball tracking using computer vision
 - Edge detection using audio + video fusion (like Snickometer + UltraEdge)
 - **Feasibility:** ⚠ Medium-Hard (video data preprocessing is heavy, but interesting).
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♦ 5. Fan Engagement & Sentiment Analysis

- **Idea:** NLP on social media (Twitter, Reddit, YouTube) for cricket events.
 - **Data Sources:** Twitter API, Reddit cricket communities, YouTube comments.
 - **Research Topics:**
 - Sentiment analysis during IPL matches
 - Popularity of players based on social buzz
 - Predicting viewership/attendance using social trends
 - **Feasibility:** ✅ Easy (text data is abundant, NLP is well-supported).
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♦ 6. Strategy & Optimization Research

- **Idea:** Game-theoretic or AI-based strategies.
 - **Data Sources:** Ball-by-ball data (Cricmetric, Cricsheet).
 - **Research Topics:**
 - Optimal batting order selection
 - Bowling change strategy optimization
 - Resource allocation in T20 matches (similar to Duckworth-Lewis-Stern but AI-based).
 - **Feasibility:** ✅ Good for algorithmic/optimization-focused M.Tech projects.
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✅ **Recommendation for M.Tech Student (easy + impactful + available data):**

1. **Match Outcome Prediction (ML/DL)** – easiest and data-rich.
2. **Player Performance Analysis (ML + Stats)** – practical and feasible.
3. **Sentiment Analysis of Cricket (NLP)** – lots of data, quick to implement.

Would you like me to **list specific project titles** (with scope + dataset links) so you can directly pick one for your M.Tech research?



list specific project titles (with scope + dataset links + previous research) so you can directly pick one for your M.Tech research?

Here are **seven concrete and feasible M.Tech project ideas** in cricket analytics—each includes a clear title, project scope, recommended datasets (easily accessible), and references to previous research for grounding and inspiration:

1. Ball-by-Ball Match Outcome Prediction Using ML/DL

- **Scope:** Build a machine learning or deep learning model that uses ball-by-ball data to dynamically predict match outcomes—updating probabilities as the match progresses.
- **Datasets:**
 - **Cricsheet:** Provides free ball-by-ball data for T20 internationals, ODIs, IPL, etc. (used in GitHub projects) [GitHub](#)[Opendatabay](#).
 - **Opendatabay:** Full ball-level CSV suited for match-level and player-level analytics [Opendatabay](#).
- **Prior Research:**
 - *Dynamic cricket match outcome prediction* (2021): Used logistic regression, ensemble methods, and LSTM/GRU models with accuracy $\approx 76\%$ ball-by-ball [Zendy](#)[OUCI](#).
 - *Analyzing LSTM models for outcome prediction via ball-level data* (2020): LSTM models for win probability per ball [arXiv](#).
- **Why it works:** High-impact, data readily available, and techniques from the literature can be directly applied or extended.

2. Pre-Match Winner Prediction in T20 Using ML

- **Scope:** Develop a model that predicts the match winner before the first ball, leveraging team statistics, past match-ups, venue, and ranking.
- **Datasets:** Extract features from **ESPN Cricinfo** (e.g., team form, venue performance). Use ball-by-ball archives to create aggregated features.
- **Prior Research:**

- *Forecasting T20 match winners through ML (2024)*: Compared several models (Logistic Regression, Random Forest, XGBoost); Random Forest achieved ~84% accuracy [OUCI](#).
 - **Why it works**: Simpler data prep, great baseline accuracy, good for teams and coaching strategies.
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3. Context-Aware Player Performance Metric (CAMP)

- **Scope**: Implement a context-aware performance metric that rates players based on match situations (opponent strength, pressure scenarios, etc.). Evaluate how well this matches expert review like Man of the Match.
 - **Datasets**: Ball- or match-level data (2001–2019) from public archives or aggregated from Cricinfo/Kaggle.
 - **Prior Research**:
 - *CAMP: Context-Aware Cricket Players Performance Metric (2023)*: Outperformed Duckworth–Lewis–Stern in aligning with expert picks (83% match to MoM) [arXiv](#).
 - **Why it works**: Novel, well-motivated, and directly builds on an empirical method showing strong results.
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4. Cricket Match Sentiment Analysis for Outcome Prediction

- **Scope**: Analyze tweets around match time to extract fan sentiment and evaluate if sentiment trends can help predict match outcomes.
 - **Datasets**:
 - Public Twitter API or Kaggle’s “Cricket Tweets” dataset (~51,862 tweets) [GitHub](#).
 - **Prior Research**:
 - *Exploring Twitter Sentiments for Predicting Match Outcomes (2024)*: Evaluated sentiment features, tweet volume/timing, and their influence on predictive accuracy [ACM Digital Library](#).
 - **Why it works**: Abundant text data, high novelty, suitable for NLP-oriented project.
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5. Player Strength and Weakness Profiling via Text Commentary

- **Scope**: Parse large volumes of ball-by-ball or summary text commentary to detect patterns of player strengths or weaknesses, phrased as rules or behavioral profiles.
 - **Datasets**: Commentary archives from ESPN Cricinfo, Kaggle, or previously collected datasets.
 - **Prior Research**:
 - *Cricket Player Profiling using Text Commentary (2023)*: Analyzed 1 million+ commentaries, derived strength/weakness rules, and validated against known performance profiles [arXiv](#).
 - *Impact of a Batter in ODI via Regression from Commentary (2023)*: Created “Effective Runs” metric from commentary and match context using regression models (~90% accuracy) [arXiv](#).
 - **Why it works**: Unstructured data offers rich, underutilized information—great for NLP + rule mining.
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6. Impact of Protective Gear on Player Agility

- **Scope:** Analyze how wearing protective gear (pads, gloves) affects agility and sprint performance using empirical measurements. You could model performance trade-offs and suggest gear configurations.
- **Datasets:**
 - Mendeley Data release (2025): Contains agility/sprint test results (without and with gear) for 72 players across age groups, plus demographics and performance times [Mendeley Data](#).
- **Why it works:** Unique dataset with physical performance metrics; ideal for combining sports science and ML/regression modeling.

7. Video-Based Ball Delivery Detection & Tracking (CV)

- **Scope:** Build a computer vision pipeline to detect and segment ball delivery frames in cricket videos; optionally track ball trajectory or classify ball pitch length.
- **Datasets:** YouTube cricket clips, annotated frames, or recently released segmentation datasets (~298 frames, 9 classes) [Reddit](#).
- **Prior Research:**
 - *Computer Vision Approach for Segmentation and Tracking of Ball Deliveries* (2022): Used MobileNet and YOLO for ball delivery segmentation and tracking bounding boxes [arXiv](#).
- **Why it works:** Combines video data and Deep Learning; suitable for CV-focused M.Tech student.

Summary Table

#	Project Title	Data Source	Key Reference
1	Ball-by-Ball Match Outcome Prediction	Cricsheet, Opendatabay	Goel et al. (2021), Chakwate & R.A. (2020)
2	Pre-Match Winner Prediction – T20	ESPN Cricinfo	Chakraborty et al. (2024)
3	CAMP – Context-Aware Metric	Match archives	Ayub et al. (2023)
4	Sentiment Analysis from Twitter	Twitter API, Kaggle	Procedia CS (2024)
5	Profiling via Commentary NLP	Commentary archives	Behera & Saradhi (2023); Al Asad et al. (2023)
6	Protective Gear vs Agility	Mendeley dataset	Boby et al. (2025)
7	CV for Ball Tracking	Video frames, annotated datasets	Abbas et al. (2022)

Suggestions for Getting Started

1. **Pick one theme:** Consider your interest—ML, NLP, CV, sports science, or hybrid—and choose the project aligned with that.

2. **Acquire and explore data:** Download your selected dataset (Cricsheet, Kaggle, Mendeley, commentary archives).
 3. **Survey related work:** Read cited papers to understand methodology and benchmarks.
 4. **Define your innovation:** For example, extend accuracy, real-time prediction, richer features, or apply to women's cricket.
 5. **Plan deliverables:** Dataset pipeline, modeling, evaluation (accuracy, F1, etc.), visualization/dashboard, documentation.
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Let me know which project intrigues you most—I can help flesh out detailed scope, milestones, evaluation metrics, or even suggest thesis titles!